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B2auth: A contextual fine-grained behavioral biometric authentication framework for real-world deployment

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ABSTRACT

Several behavioral biometric authentication frameworks have been proposed to authenticate smartphone users based on the analysis of sensors and services. These authentication frameworks verify the user identity by extracting a set of behavioral traits such as touch, sensors and keystroke dynamics, and use machine learning and deep learning techniques to develop the authentication models. Unfortunately, it is not clear how these frameworks perform in the real world deployment and most of the experiments in the literature have been conducted with cooperative users in a controlled environment. In this paper, we present a novel behavioral biometric authentication framework, called B2auth, designed specifically for smartphone users. The framework leverages raw data collected from touchscreen on smartphone to extract behavioral traits for authentication. A Multilayer Perceptron (MLP) neural network is employed to develop authentication models. Unlike many existing experiments conducted in controlled environments with cooperative users, we focused on real-world deployment scenarios, collecting data from 60 participants using smartphones in an uncontrolled environment. The framework achieves promising results in differentiating the legitimate owner and an attacker across various app contexts, showcasing its potential in practical use cases. By utilizing minimalist data collection and cloud-based model generation, the B2auth framework offers an efficient and effective approach to behavioral biometric authentication for smartphones.

1. Introduction

With the diversity of sensors and services on smartphone different behavioral biometric authentication models have been developed and achieved high performance results. Unfortunately, the majority of these developed models are facing serious challenges that need to be addressed [1,2]: (1) the collected data from the smartphone sensors are suffering from some level of impreciseness (i.e., noisy data). Consequently, we need to clean these sensory data by collecting a more fine-grained information that depicts the exact behavior of the user [3,4]; (2) the majority of developed authentication models were trained on data that were collected in a single app-context. To solve this issue, the models should be trained on data collected from different app-contexts; (3) Deep learning has increased the capabilities of mobile applications. However, one of the crucial challenges that confront the utilization of deep learning on mobile devices, is the limited computational power available. A significant amount of computation is

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needed to process huge amounts of data at training and prediction. Furthermore, the amount of memories in smartphones are smaller compared to personal computers, so the memory consumption should be considered during the design of an active authentication systems.

In this paper we develop a contextual fine-grained behavioral biometric authentication framework called B2auth, in which a minimalist data collection and feature extraction approaches were conducted on the device and the model generation on the cloud. The framework collects data from the touchscreen and performs features extraction process on the smartphones to capture discriminative temporal and spatial attributes that distinguish the legitimate user from imposters. The extracted features are then transmitted to the cloud to train a deep learning model, enabling identification of the current user of the smartphone. The cloud-based model is subsequently exported to the mobile device for on-device inference (i.e., the process of making predictions on the smartphone), offering several advantages. These include offline prediction capability, as there is no need to send the data to the cloud and wait for results, thus avoiding latency. The generated model can be updated to adapt to the changes in user's behavior, ensuring ongoing accuracy and robustness in the authentication process.

The novelty of B2auth lies in its contextual fine-grained approach to behavioral biometric authentication, which incorporates a diverse set of behavioral traits extract from the touchscreen. Unlike traditional laboratory-based studies, B2auth conducted a real-world field study with 60 participants, collecting a substantial dataset of 34,621 touch events in uncontrolled environments, reflecting authentic user behaviors. The framework's efficient feature engineering module enables real-time extraction of distinctive features, reducing computational burden and making it suitable for resource-constrained devices like smartphones. Moreover, B2auth has been deployed in real-world scenarios, showcasing its practicality and effectiveness in authenticating users based on their behavioral patterns. The validation on real-life smartphone samples further demonstrates its applicability, emphasizing its potential to provide secure and seamless user verification in practical settings, making it a promising solution for real-world deployment. Importantly, the study also employed a custom test-bed, purpose-built to ensure rigorous evaluation and validation of B2auth's performance in real-world deployment scenarios.

The contributions of this paper can be summarized as follows:

- We developed B2auth a contextual fine-grained behavioral biometric authentication framework.
- We conducted a real-world field study over 60 participants who used the smartphones without any intervention. Consequently, a dataset of 34621 touch events were formulated of actual users behaviors over different applications context. The collected data reflects the realistic and the diversity of usage behavior of users.
- We evaluated the B2auth framework by conducting an authentication study. In which we measured the applicability of the performance of the generated authentication models for the enrolled users.
- We validate our framework on real world deployment scenario on smartphones, showcasing that the system can achieve robust authentication using only a small number of touch actions, enhancing usability and efficiency.

The rest of the paper is organized as follows: Section 2 presents the background and discusses the related work. The threat model is presented in Section 3 followed by a details description about the framework modules in Section 4. A field study is presented in Section 5. Then, we provide the performance evaluation and the results of the B2auth framework in Section 6. Section 7 provides a detailed discussion about B2auth. Finally, Section 8 provides the conclusion and future work.

2. Background and related work

Behavioral biometric authentication is considered as an automated recognition process that verify the claimed identity by analyzing the usage behavior [5,6]. The primary objective of this process is to protect the device against the unauthorized access. Two main characteristics serve as the foundation of any behavioral biometric authentication system: continuity and implicitness [3,7].

Behavioral biometric authentication relies on specific biometric traits touch gesture, keystroke dynamics or behavioral sensor profiling. Table 1 shows a summary of state-of-the-art studies conducted to develop behavioral biometric authentication frameworks. The majority of these studies were conducted in a controlled environment using a custom-built apps [8,9].

Researchers have explored touch gesture behaviors as unique biometric traits to authenticate smartphone users. For instance, Song et al. [10] introduced a multi-user authentication system for smartphones, leveraging touch gesture behaviors for continuous user identification. Similarly, Peng et al. [13] conducted an experimental study, proposing touch biometrics for continuous user authentication on smartphones. Cai et al. [18] compared various touch-based user identification techniques to assess their effectiveness. Meng et al. [21] focused on enabling touch-based authentication for wearable devices on different materials. Syed et al. [25] explored the impact of three key factors on the authentication performance of a touch-based authentication system: user posture, device size, and device configuration. Through rigorous experimentation and analysis, the study seeks to uncover how variations in user posture, different device sizes, and configurations can influence the accuracy and reliability of the touch-based authentication process. These studies collectively demonstrate the potential of touch-based biometrics in providing seamless and secure user authentication for mobile devices, enhancing overall user experience and device security.

The sensor-based approaches involves utilizing various sensors in a smartphone to capture and analyze unique user behaviors. These sensors include accelerometers, gyroscopes, magnetometers, ambient light sensors and proximity sensors. Sensor-based behavioral biometrics have been investigated in several studies. For instance, Lee et al. [11] introduced an implicit authentication system for smartphone users, utilizing their unique behavioral characteristics and leveraging the integrated sensors into smartphones. The proposed system introduces a context-based authentication models, enabling the discrimination between the legitimate owner

Table 1

This table contains the-state-of-the-art studies for designing behavioral biometric authentication.

Study	# Users	Dataset (# Samples)	Biometric traits	# features	AI model	Performance(%)
Song et al. [10]	161	Private(15000)	touch gesture	36	KNN,SVM	EER:1.88
Lee et al. [11]	35	Private(800)	sensor-based	14(smartphone)	RF, KRR	FRR:0.9,FAR:2.8
Muaaz et al. [12]	35	Private(272 m)	gait	N/A	DTW	EER:13.0
Peng et al. [13]	32	Private(-)	touch gestures and voice	45	OCSVM	EER:9-16.6
Amini et al. [14]	47	Private	sensor-based	TDF and FDF	LSTM	Acc: 96.7%
Buriro et al. [15]	97	Private (10500)	sensor-based	38	BN, RF, MLP, KNN	TAR:85%
Krishnamoorthy et al. [16]	94	Private(-)	keystroke dynamics	36	SVM	accuracy:97.4
Centeno et al. [17]	90	Public(6750/user)	sensor-based	N/A	Siamese CNN,OCSVM	98
Cai et al. [18]	20	Private(2948)	touch gestures	16	AROW,NHERD,SCW	EER<4.0
Shen et al. [19]	102	Private(520,200)	touch gestures and sensor	38	HMM,SVM,NN	EER:4.71
Shen et al. [20]	10	Private(27,681)	sensor-based	37	ten one-class detectors	EER:2.21
Meng et al. [21]	48	Private(-)	touch gestures	21	PSO-RBFN	average EER:2.4
Valero et al. [22]	50	Private(-)	Sensors+application	28	Isolation Forest	Rec:0.92, Pre:0.77
Acien et al. [23]	48	Public(-)	touch+sensors+BP	N/A	SVM	ACC:82.2-97.1
Smith-Creasey et al. [24]	20	Private(18000)	touch gestures and sensor	6 groupings	LR,NB,RF	EER:0.81
Buriro et al. [9]	85	Private(10,200)	touch gesture and sensor	128	RF	TAR:99.35
Syed et al. [25]	31	Private(13080/user)	touch gestures	14	SVM,LR,NB,RF,MLP	EER:5.4
Yang et al. [8]	45	Private(250,000)	touch gesture	10	OCSVM,iForest	accuracy:95.85
Zou et al. [26]	118	Private(74,142)	gait	NA	DCNN, LSTM	accuracy:93.7
Shi et al. [27]	77	Private(14,850)	gesture and sensors	NA	OCSVM	EER:5.1139%
Zhihao et al. [28]	100	Private(30 Million)	Gesture and sensors	up to 144	DeSVDD	EER:8.814.9%
Shen et al. [29]	61	Private(200000)	Gesture, Sensors, Context	3 groupings	GBDTNN	Accuracy 97.4%

and other users of smartphone. Amini et al. [14] developed DeepAuth, an LSTM model that uses sensor data to authenticate smartphone users based on their behavioral patterns. Buriro et al. [15] explored touch-tokens as an implicit modality for mobile user authentication, using sensor-based data as part of the authentication process. Centeno et al. [17] employed an autoencoder based approach to authenticate users based on behavioral biometrics extracted from smartphone sensors.

A combination between different modalities was introduced by Shen et al. [29], where they developed a continuous authentication approach using touch, sensor and context behavior biometrics. The key idea lies in its context-aware feature set, which adeptly characterizes touch operations in usage contexts, overcoming limitations observed in conventional representation methods. The sensor-based approach offers advantages such as being less intrusive and unobtrusive, as it does not require explicit user actions for authentication. It can continuously and passively verify users in the background, enhancing the overall user experience. By leveraging a combination of sensor data, these studies contribute to the development of secure and user-friendly authentication systems, advancing the field of behavioral biometrics.

The keystroke dynamics-based approach is another prominent method in behavioral biometric authentication. For instance, Krishnamoorthy et al. [16] conducted a study focused on identifying smartphone users through their unique keystroke patterns, showcasing the feasibility of this approach. Additionally, some experiments in the literature combined keystroke dynamics with other traits for multi-modal authentication. For instance, Peng et al. [13] proposed a continuous user authentication system using touch biometrics and voice, demonstrating the potential of combining these factors to improve authentication accuracy. Chauhan et al. [30] proposed a behavioral-based authentication system that combined deep and online learning models for incremental and adaptive learning. Their system was evaluated on various behavioral modalities such as breathing, gait and EMG, showing promising results in maintaining high TPR while minimizing FPR.

Although classical machine learning models like KNN, SVM, and RF were commonly used in these studies, there is a notable scarcity of deep learning models applied to keystroke dynamics-based authentication [10,11,13].

3. Threat model

We consider an adversarial threat model scenario in which an attacker has physical access to the smartphone. The attacks might come from either stranger who steals the smartphone or a family member, close friend or co-worker [31]. If the attacker can unlock the device through obtaining the credentials, this results in gaining access to the device and sensitive information will be available to the attacker including user's applications, social accounts, and personal data [32]. Our main focus is to develop an active smartphone authentication mechanism by analyzing touch gestures in an unobservable, continuous and transparent manner [7]. Therefore, the attacker will be unmindful that she is in middle of implicit identity verification while interacting with the smartphone, and consequently the results will be reported and a corresponding actions will be performed. It is worth mentioning that we primarily consider a verification process of recognizing the user by performing one-to-one matching and validates the claimed identity of a user against the smartphone owner, as a smartphone is usually a private property and mostly not shared by others.

4. B2auth framework architecture

In this section we provide a comprehensive description about the architecture of B2auth. Fig. 1 shows the main components of the architecture supported with the data flow between these components. The first component of B2auth operates on the device (i.e., the lower part in Fig. 1) and contains the instrumentation process, data collection, preprocessing, feature construction, and the verification phase. Then, the device-based component is responsible for capturing user behavioral data, extracting relevant

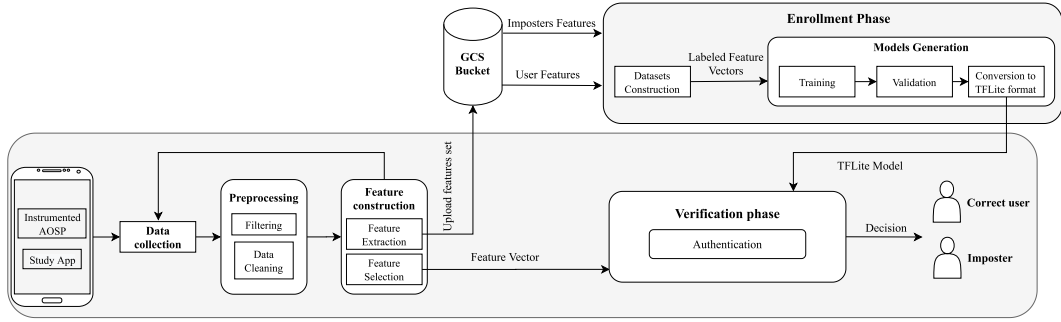


Fig. 1. B2auth Architecture.

features, and performing the initial verification process locally on the mobile device. The second component (i.e., the upper part in Fig. 1) operates on the cloud and contains dataset construction, model training, validation and model generation. The cloud-based component utilizes the data collected from multiple devices to construct a comprehensive dataset, train a deep learning model, validate its performance, and generate the final authentication model. The trained model is then exported to the mobile device, enabling on-device inference for efficient and offline user authentication. The combination of these two components ensures a robust, scalable, and adaptive behavioral biometric authentication system.

4.1. Real-time testbed for B2auth

To enable the real-time functionality for our implicit authentication framework, we established a testbed using the LineageOS Android distribution (i.e., AOSP Pie, API level 28) [33]. The testbed involved the instrumentation of the LineageOS distribution, which included the development of two new classes, Stroke and TouchEvent, designed to represent the building blocks of touch gestures and feature extraction process. Furthermore, we instrumented the View and ViewGroup classes, which are part of the AOSP framework base layer, to incorporate touch events data collection based on different criteria, as described in Section 4.2. This testbed provides a robust platform for evaluating the performance and usability of B2auth in a dynamic, interactive mobile environment.

4.2. Preprocessing

Touch gesture occurs when the user places his finger on the screen. Consequently, the data collection of the framework starts collecting data once the user touches the screen. These data come in a sequence of touch events fired in several times. Unfortunately, the sequence that represents the gesture contains repeated touch events, and sometimes the last event that represents the end of the gesture is missing which makes it hard to detect that gesture. To handle these issues, we developed a set of filtering criteria to be performed during the data collection to get a cleaned gesture which accurately represents the user touches. The main criterion in the filtering process is to remove the duplicated touch events to avoid redundancy and represent the actually touches on the screen. Then, we categorized touch gestures in two types, tap and slide. For tap gestures which represent touches conducted in a small part of the screen. These tap gestures occurred when a press action is done such as button press or option selection. On the other hand, slides gestures which represent the touches that were conducted in large parts of the screen. These slides gestures happened to do actions such as scrolling or moving finger.

4.3. Feature engineering

Feature engineering involves extracting meaningful information from raw touch data to create informative features for subsequent analysis and model training. The extracted features sets are shown in Table 2 and have been used to build two different authentication models, one for slides and another for taps.

The first three feature sets—X, Y Co-ordinates, Pressure, and Area—are directly obtained from smartphones, likely representing Cartesian positions during touch gestures, applied touch pressure, and the size of the touched area, respectively.¹ The subsequent feature sets are obtained using a predefined calculation. Length and Displacement involve spatial measurements based on changes in X and Y positions. Duration captures the elapsed time between touch-down and touch-up events, while Inter-stroke-duration measures the time between consecutive touch gestures. Velocity is computed as the ratio of Displacement to Duration, representing the speed of the touch and is calculated using the following equation:

$$V = \sqrt{((X')^2 + (Y')^2)}, \quad (1)$$

¹ The touch pressure is auto-calculated on smartphones using capacitive touchscreen sensing. So the value of the touch pressure is approximately represent the physical pressure applied to the touch device and its value is normalized between 0.0 (no touch) and 1.0 (full force).

Table 2
The extracted features list with a brief description.

Features set	Brief description	# Type
X,Y co-ordinates	Co-ordinates of up and down touch position	4 Numeric
Pressure	Up and down touch pressure	5 Numeric
Area	Size of up and down touched area	5 Numeric
Length	Total distance of touch gesture	1 Numeric
Displacement	The direct distance between touch up and down	1 Numeric
Duration	Time duration of whole touch gesture distance	1 Numeric
Inter-stroke-duration	The duration between previous and next gesture	1 Numeric
Velocity	The speed of performing a touch gesture	2 Numeric
Curvature	The amount of deviation of the touch from being a straight	1 Numeric
Acceleration	Average acceleration of touch gesture	1 Numeric

where X and Y are sets of touchpoints in the touch gesture and V represents the velocities vector at each touchpoint. Curvature K represents the deviation amount of the touch gesture trajectory from a straight line and is computed using the following formula [34]:

$$K = \frac{|X'Y'' - Y'X''|}{(X'^2 + Y'^2)^{\frac{3}{2}}}, \quad (2)$$

where X and Y represent the vectors of the touchpoints in the touch gesture. Their primes refer to the first and second order of derivatives. Acceleration represents the average acceleration during the touch gesture and is calculated using the following equation [35]:

$$A = \frac{d^2s}{dt^2}T + k\left(\frac{ds}{dt}\right)^2N \quad (3)$$

Here, s represents the touchpoints and, d denotes the derivatives with respect to the time, and T and N represent the tangent and normal vectors, respectively. These features collectively provide a deep understanding of smartphone interactions, encompassing spatial, temporal, and dynamic characteristics.

Having too many features can cause the model to capture unimportant patterns and learn from noise [36]. In addition, it can slow down the training process and cause the model's inference to be slower. To reduce the number of features and eliminate redundant or irrelevant features, we conducted a feature selection process. In this process we excluded the least contributed features based on feature importance analysis [37]. This helped in identifying the most helpful group of features and narrowing the set of features to those most relevant to the AI models. Fewer redundant features means fewer chances of making decisions based on noise which reduces overfitting. It also improves the accuracy and optimizes the model performance by decreasing training time and accelerating inference. For instance, we excluded the *Inter-stroke-duration* from the slide features. In case of tap features, we excluded all features related to touch area, in addition to length, displacement, velocity, curvature and acceleration because the touch area of the tap is very tiny. [Appendix C](#) contains a detailed list of selected features.

4.4. Enrolment phase

4.4.1. Dataset construction and labeling

The purpose of data construction and labeling is to prepare the dataset needed for training process (i.e., Section 4.4.3). The B2auth framework used the Google AI platform which provides training and prediction services for developers to help them train their models and deploy them via the Google cloud platform (GCP) [38]. A training application was developed to make use of those services. When the framework submits a training job to the AI platform during the enrolment phase, the training application initially downloads the features of the legitimate user as well as the imposters from the Google Cloud Storage Bucket (GCSB) to the AI platform instance.

The features of legitimate (i.e., positive samples) user were collected during the B2auth study described in Section 5. For imposter features, a separate set of participants engaged in the enrolment scenario, playing the role of impostors (i.e., negative samples). Importantly, these participants did not contribute to the primary study. This separation of negative and positive samples was a methodological choice, aiming to prevent any mixing that could introduce bias or compromise the integrity of the study.

The B2auth framework uses two deep learning models (i.e., Section 4.4.3), namely classifiers, that were trained on data from both the legitimate users and imposters. The two models calculate authentication scores that represent the probabilities of the current user being a legitimate user. The first model is trained on the slides dataset, and the second model is trained on the taps dataset. In regard to the construction of the slides dataset, the slides features of imposters are initially shuffled, then a certain number of samples are drawn from them to be concatenated with the slides features of legitimate user. Likewise, the taps features of legitimate user and the taps features of imposters are used to construct the taps dataset.

4.4.2. Training, validation and testing

In regard to preparing the datasets for training the deep learning models, each dataset goes through the following procedure to be pre-processed properly. First, the features and labels are separated, and then the dataset is split into training, validation and test sets by 70%, 10% and 20% respectively. A feature scaling technique is performed using the standardization method to bring the distribution of values of each feature to zero mean and unit variance. The training set as well as the validation set are used during training to handle the under-fitting or overfitting of the model. Once the training process ends, the test set is utilized to evaluate the model's performance on the unseen data to determine how well it generalizes to data it has not seen before.

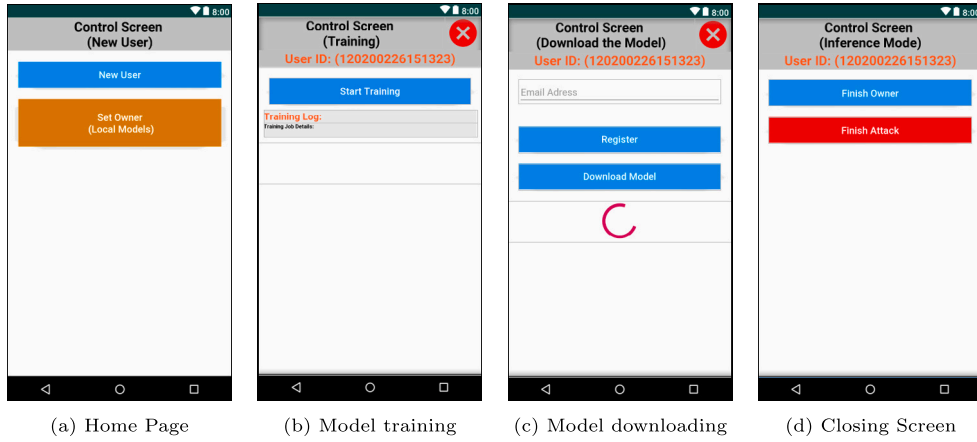


Fig. 2. Samples of screenshots of the study app.

4.4.3. Model generation

We design a Multilayer Perceptron (MLP) neural network that comprises three hidden layers, each followed by batch normalization layers, and an output layer with a single neuron. Employing rectified linear unit (RELU) activation functions for the hidden layers and a sigmoid activation function for the output layer, the model takes an appropriate feature vector as input and produces a real number within the range of 0 to 1. This output represents the probability that the given feature vector belongs to the legitimate user class. The network was intentionally kept simple to minimize on-device inference costs when running concurrently with other applications.

Hyper-parameters tuning was conducted using Bayesian optimization, supplemented by additional manual adjustments, as outlined in [39]. To enhance convergence speed, the learning rate is dynamically reduced if no performance improvement is observed over a specific number of epochs. Additionally, the early stopping technique was implemented to mitigate overfitting and streamline the training process.

In order to evaluate the performance of the model, the receiver operating characteristic (ROC) curve and the area under curve (AUC) were used as evaluation metrics. When the training ends, the performance of the model on unseen data is evaluated using the held-out test set to estimate the generalization error. Moreover, the values of the evaluation metrics on the test set are fetched and uploaded to the GCSB for each user to be checked later when needed. Afterwards, the model is saved and converted into smaller and more efficient tensorflow lite (i.e., TFlite) format to be deployed on the smartphone. This format is optimized for on-device inference through hardware acceleration and model optimization. It also supports multiple platforms covering Android, IOS and embedded devices [40].

4.4.4. Model deployment

Once the conversion to TFlite format is done, the slides-based model is uploaded to a GCSB so that it can be fetched and utilized by the B2auth framework in the smartphone. Then, GCP sends a notification of completion to the B2auth framework so that the verification phase can start. Similarly, the previous procedure is applied to the taps dataset to produce the taps model.

4.5. Verification phase

In the verification phase, the collected features are fed directly to the trained model to compute the authentication scores $P_1(u)$ and $P_2(u)$ that represent the probability of the current user being a legitimate user based on the slides features and the taps features, respectively.

5. B2auth study

5.1. Study app

We developed an android application to help in running the B2auth field study. The app has two modes, the training mode, and the authentication mode. In the training mode, where the study app acquires the biometric data, analyzes these data and extracts a distinctive set of features, then it builds the feature templates and upload it to the GCP buckets for training [38]. Similarly, in the authentication mode, the study app does the same process but instead of uploading it to the GCP, it sends the extracted features directly to the model for authentication.

As shown in Fig. 2, the study app is responsible for managing the data collection, feature extraction and the interaction between the device and the GCP. This interaction includes uploading the features to the GCSB, starting and tracking a training job, downloading the trained models and finally adding these models locally to the device storage to be used for authentication.

The following points briefly describe the interaction between the study app and the components of the B2auth framework in the enrolment mode:

1. **Data collection:** we instrumented the AOSP framework layer with a logging tool that listen for the finger movement events on the touchscreen for any foreground app (i.e., see Section 4.1). The raw data related to touch gesture such as x and y coordinates, pressure and area were collected and stored privately in the internal storage after applying preprocessing steps as described in 4.2.
2. **Feature extraction:** we instrumented the AOSP framework layer with a feature extraction module that enables B2auth to analyze the collected data and extracts a distinctive set of features, Section 4.3 shows a detailed description about the extraction process that relies on the geometric analysis of the stroke and its motion dynamics. Then, the features are aggregated to formulate a feature template. Table 2 shows a list of the extracted features with a brief description in addition to the feature type.
3. **Model Generation and Deployment:** the study app uploads the feature template to GCP and submits a training job to train an AI model using this feature template as a training set, see Section 4.4.2 for more details. Then, the AI Platform training service generates a trained model for that feature templates (i.e., Section 4.4.3 provides a detailed description). The generated model is saved in a TFlite format for deployment (i.e., Section 4.4.4 provides a detailed description).

In authentication (i.e., verification) mode, the data collection and feature extraction steps are working in the same manner as in the training mode. But instead of building a feature template, B2auth sends it instantly to the trained model for authentication. The trained model is fetched from the GCSP and integrated with the study app to perform the authentication instantly and in a real time once the user performs any touch events.

5.2. Participants

We recruited participants from our University campus through an advertisement campaign. Participants who accepted to contribute to the study were asked to answer a short survey about demographic and read a description about the study procedure to be aware of the collected data. The identities of all participants remain anonymous. All data are stored confidentially. B2auth framework uses internal storage of the device for the data storing purpose which prevents other applications' access. The collected data were kept in a secure server in our University and only accessible by the research team who has permission to access these data. All data from individual participants were analyzed and their anonymity were protected in any research papers, presentations and reports that result from this work. To protect user privacy, we obtained the approval from our university research ethics board before collecting the data. We inform the participants about the experiment and the collected data, and each participant had to accept for the app to continue. The study has been filed and approved by the vice president for postgraduate studies and research (i.e., Institutional Review Board (IRB)) and the consent form can be found in the appendix.

We were able to collect data from 63 participants who agreed to participate in the study. We excluded 3 participants who have a huge gap in the enrolment scenario. The details about demographic are shown in Table 3. The table illustrates the distribution of the participants over different parameters and properties. The study was conducted on a variety of participants including 40 (63%) males and 23 (37%) females of different ages ranged from 21 to 65 years old (mean = 26.5, median=23). Also, all participants were asked about the type of authentications they use such as PIN code, password, pattern, face, fingerprint or none. Participants can select one or more than one options according to what they use on their phones. The data collected shows that around 90% of the participants use at least one of the authentication options. It shows that around 10% out of 60 participants did not use any authentication type on their phones. Therefore, we collected the reason for using authentication as well as the reason for not using authentication. The collected data indicates that more than 50% of participants share their phones with others at least once in a daily bases, and only around 18% of the participants never share their phones with others. Also, some additional data were collected such as the participants' thoughts on authentication and from whom they want to protect their phones. Around 60% of the participants believe that using authentication will help them in protecting their phones from unauthorized access. On the other hand around 30% considered the authentication as annoying and time consuming process.

5.3. Study procedure

We invited the selected participants and gave them a Google Pixel XL device instrumented with the B2auth framework and the study app. They were asked to use it and do actions based on a predefined scenario described in Section 5.4. During the first usage, the user has to finish the enrolment phase by performing the scenario and then the device will be ready for the authentication phase after downloading the trained model. Then, the status of the B2auth framework was converted to the verification process to recognize the user by performing one-to-one matching and validates the claimed identity based on our developed AI model as described in Section 4.4.3. A detailed description about the study procedure is shown in Appendix A.

5.4. Enrolment scenario

We developed a set of common daily activities and asked the participants to do them in the field study, the detailed descriptions are shown in Appendix B. The user will use the smartphone for a while to do the enrolment scenario and then he will start the training process and the model will be generated. Once these processes are completed, the device mode will be converted from the training mode to the authentication mode. Hence the user can start using the phone again for inference session. According to the previous work, this methodology represents the best performance scenario among three other scenarios that were discussed by Shen et al. [41].

Table 3
Participants demographics, N = 60.

Parameter	Property	# of participants
Gender	Males	38 (63%)
	Females	22 (37%)
Age	19–24	38
	25–30	13
	31–40	5
	41–65	4
Education	Master or PhD	10
	University (Bachelor's)	11
	High School	39
Job	Student	39
	Researcher or staff	17
	Employee	2
	Unemployed	2
Current Authentication	None	6 (9.52%)
	PIN	20 (31.75%)
	Password	14 (22.22%)
	Pattern	20 (31.75%)
	Face	8 (12.70%)
	Fingerprint	27 (42.86%)
Sharing	Occasionally	21 (33.33%)
	Rarely	18 (28.57%)
	Daily	13 (20.63%)
	Never	11 (17.46%)
Reasons for not using authentication	It is a hassle/takes time	6 (100%)
	Nothing to hide	2 (33.3%)
	Others can use my phone in emergency	1 (16.6%)
Reasons for using authentication	Protected if lost/stolen	46 (80.70%)
	Protected when unattended	22 (38.60%)
	Someone casually picking it	19 (33.33%)
	Phone protected if misplaced	18 (31.58%)
	Unwanted disclosures	15 (26.32%)
Thoughts on Authentication	It is protected from Unauthorized access	34 (59.6%)
	It is easy	27 (47.3%)
	It is annoying	11 (19.3%)
	It takes time	6 (10.5%)
Protecting From	Strangers	44 (77.2%)
	Children	27 (47.4%)
	Friends	19 (33.3%)
	Coworkers	10 (17.5%)
	Roommate	10 (17.5%)
	Spouse	3 (5.2%)

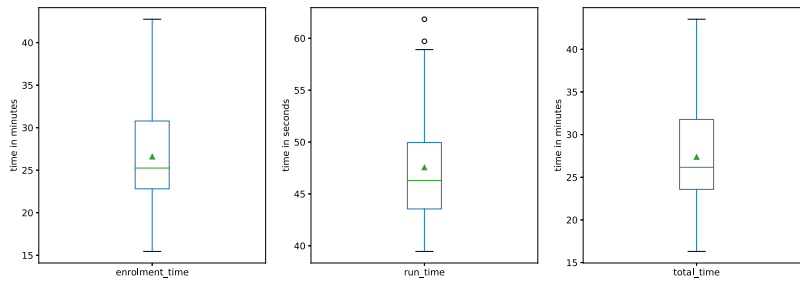


Fig. 3. Bootstrap analysis.

5.5. Authentication scenario

In the same way as in the previous section, we developed a set of commonly used actions to guide the participants to do them in the authentication session. We have chosen some commonly used apps, part of them have been used in the enrolment

Table 4

Bootstrapping time and elapsed time statistics for enrolment process. Q1, Q2 and Q3 represents the lower, median and upper quartiles.

	Elapsed time			
	Q1	Q2	Q3	Mean
enrolment_time (m)	22.8	25.2	30.7	26.5
runtime (s)	43.6	46.3	50	47.5
total_time (m)	23.6	26.2	31.8	27.3

Table 5

Analysis of touches in the enrolment session.

	Touch actions			
	Minimum	Median	Maximum	Mean
taps	150	371	889	362
slides	62	215	491	214
total touches	227	578	1150	577

phase (i.e., screen unlock, chrome, game, WhatsApp), and others were different and have not been used before in the enrolment (i.e., Youtube, Souq, crazy subway). This scenario has been conducted by the owner (i.e., authentication session).

5.6. Attack scenario

To test the performance of the generated model on unseen data, participants were asked to do an attack session in which the participant play the attacker role. The attacker did the same scenario on the same used apps as in the authentication scenario, (i.e., screen unlock, chrome, game, WhatsApp, Youtube, Souq, crazy subway) on a device shipped with the AI authentication model for another participant [42].

6. Evaluation

6.1. Performance metrics

In order to measure the performance of the developed framework, we used Receiver Operating Characteristic (ROC) curve. It depicts the trade-off between TPR and FPR over various thresholds. Also, we used Area Under the Curve (AUC) to measure the quality performance of the model as an alternative to the accuracy, as AUC is resistance to the high class imbalance. For error rate analysis, we used the False Rejection Rate (FRR) and False Acceptance Rate (FAR). The FAR represents the ratio between the number of false acceptance and the total imposter samples while the FRR represents the ratio between the number of false rejection and the number of samples of legitimate users.

6.2. Evaluation dataset

We formulated a dataset from the collected data that was described in B2auth study in Section 5. The dataset contains features extracted from gesture actions in real time while conducting the study as described in Section 4.3. The dataset contains 34621 touch events (i.e., 21723 tap, 12898 slide) collected in the wild from 60 participants who used the phones for 26.5 min on average (i.e., MEDIAN = 25.2, SD = 5.8) performing an enrolment scenario described in Section 5.4. Participants used the smartphones without any intervention, consequently all collected events reflects the behavior of using the smartphone in a real world manner.

6.3. Bootstrap analysis

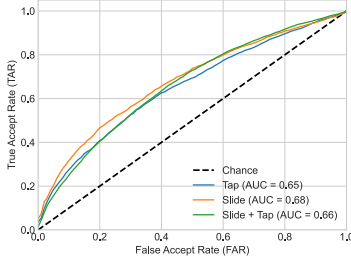
In the bootstrap analysis we measured how long does it takes to bootstrap the authentication process. To do so, we measured two durations, the time spent in the enrolment phase (*enrolment_time*) and the model generation time (*total_time*). For the enrolment time, which is calculated based on the enrolment scenario (see Section 5.4). Fig. 3 shows the quartiles of the enrolment times for participants. As we can see in that figure, the enrolment times ranged from lower quartile of 22.8 min to the upper quartile of 30.7 min with a median of 25.2 min. In case of the deployment time which represents the time it takes to generate the models. The consumed times ranged from lower quartile of 43.6 s to the upper quartile of 50 s with a median of 46.3 s. To sum up, the total time required to bootstrap the authentication is 27.3 min on average in which the user performed the aforementioned scenario 5.4 doing 577 touch actions (i.e., 362 taps and 215 slide) on average. Tables 4 and 5 shows a summary of all elapsed times and touches for all sessions.

6.4. The performance of model generation

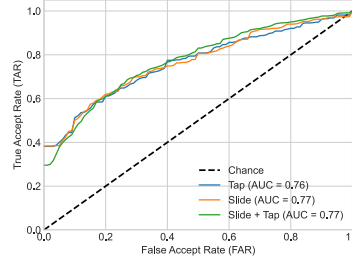
After finishing the enrolment, the collected features are uploaded to GCP bucket for training and deployment. This process was conducted for each user and the consumed time was around 47.5 s on average (i.e., MEDIAN = 46.3, SD = 5). This time includes only

Table 6
Basic results during the training on GCP.

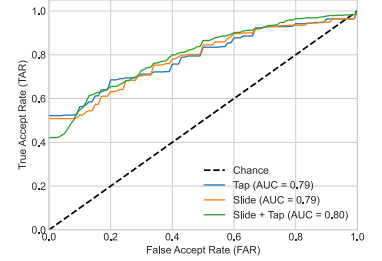
	Tap			Slide		
	Min	Max	Mean	Min	Max	Mean
AUC	0.59	0.97	0.78	0.68	0.98	0.87
FAR	0.02	0.63	0.33	0.00	0.42	0.13
FRR	0.02	0.65	0.24	0.08	0.96	0.33



(a) Confidence level over one sample.



(b) Confidence level over 10 samples



(c) Confidence level over 20 samples

Fig. 4. The confidence level of authentication over different time series.

Table 7
Authentication performance over different windows for Taps, Slides and both.

	Tap			Slide			Slide + Tap		
	w=1	w=10	w=20	w=1	w=10	w=20	w=1	w=10	w=20
AUC	0.65	0.76	0.79	0.68	0.77	0.79	0.66	0.77	0.80
EER	0.39	0.31	0.29	0.37	0.30	0.30	0.38	0.30	0.29

the training and the model deployment processes where the uploading time was excluded. Table 6 shows the average performance of generated models on GCP over all users. The average performance of slide models was 87% while the average performance of tap models was 78%. As we can see, the models generated from the slide features are performing better by almost 9% than the models generated from tap features. Some models achieved 0% of FAR which means there is no false acceptance. This reflects the high ability of the model to detect the imposters patterns. The average FAR for slide models was 13% while the FRR was 33%. This means the models have better ability of rejecting the imposters than the ability of accepting the legitimate users. This reflects the upper hand that the models have in security in compare with the usability, where the FRR represents the false rejection rate of legitimate users.

6.5. Frequency-based performance

In this section we present the authentication performance during the authentication and attack scenarios (i.e., see Sections 5.5 and 5.6). The authentication was performed instantly using a single sample, but we present the authentication performance not only using one sample but also by averaging the authentication scores over a consecutive set of samples over a specific frequency called *frequency window*. In this analysis we used three different sizes of frequency window which are one, ten and twenty. Concerning the variation of the performance of authentication with respect to the classification thresholds and the frequency windows. Fig. 4 depicts the performance of authentication via ROC curves for Tap, Slide and Tap+Slide over the three aforementioned windowing. It can be clearly seen that the performance of Tap, Slide and Tap+Slide is very close to each other where the AUC was 65%, 68% and 66% in case of using a single sample as shown in Fig. 4(a). Moreover the performance tends to increase as the number of samples in the frequency window increases. For instance, the AUC performance over window of ten samples increased almost by approximately 10% for Tap, Slide and Tap+Slide as shown in Fig. 4(b). A better performance was achieved by using a window of 20 samples, see Fig. 4(c), where the AUC performance was 79% for Tap and Slide, and it reached to 80% during a window of Tap+Slide. Similarly for the error rate analysis, where the EER decreased from 39% in case of Tap over a window of one sample to 29% in case of Tap+Slide over a window of 20 samples. Table 7 presents a summary of all results.

6.6. Time-based performance

Further analysis of the authentication performance was conducted by plotting the authentication scores predicted by the implicit authentication model over different time series. In this section, the inference of the authentication is investigated through plotting the authentication scores obtained from the slides and taps models for both the legitimate user (i.e., owner) and the

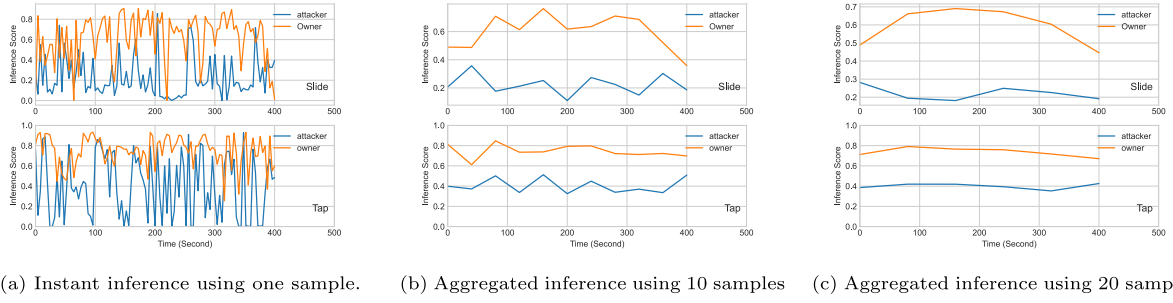


Fig. 5. The inference score of authentication over different time series. The inference score refers to the prediction generated by the model for a given touch input.

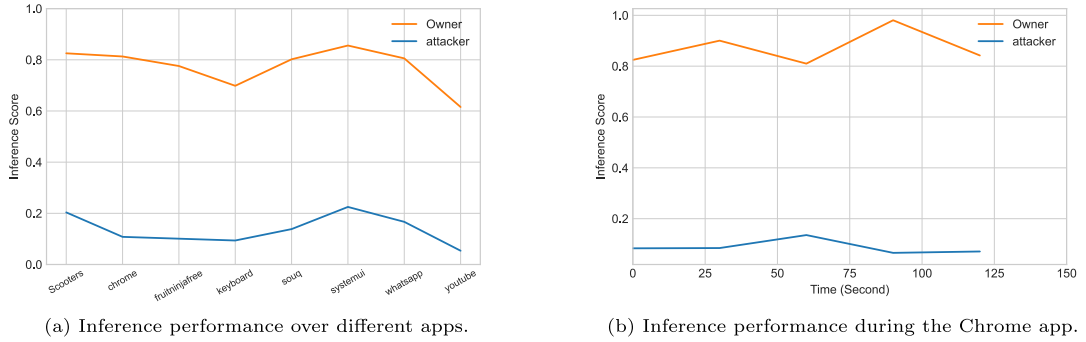


Fig. 6. The inference score of the implicit authentication model for the owner and the attacker for different app-contexts. The inference score refers to the prediction generated by the model for a given touch input.

attacker (i.e., imposter), see Fig. 5. In addition, the size of the time window was varied in order to explore the impact of the aggregation of samples on the authentication confidence. Fig. 5(a) shows the authentication inference for both slides and taps over time series when the authentication decision is done instantly based on a single sample. Furthermore, Fig. 5(b) illustrates the inference with an aggregation of 10 samples, whereas Fig. 5(c) depicts it when the window size reaches 20 samples. By observing the three plots, it can be inferred that, utilizing a bigger time window helps increase the authentication inference in verifying the claimed identity of the owner in compare with decreasing the inference of the attacker. Besides, aggregating more samples to be used in the authentication process helps the AI models to differentiate between the owner and attacker more accurately.

6.7. Contextual-based performance

To measure the performance of the developed model based on different app-contexts. Fig. 6(a) shows a clear vision of how the inference scores of the implicit authentication model varied over different apps. The inference score of a model typically refers to the prediction generated by the model for a given touch input. The x-axis contains the most commonly used apps during the enrolment and the authentication scenarios (i.e., see Sections 5.4 and 5.5 respectively). The developed model keeps the owner inference scores greater than 60% over all apps, some of them passed the 80% score such as *WhatsApp*. On the other hand, the attacker inference scores ranges between 5% to 22%. Consequently the developed implicit authentication model managed to differentiate between the inferences of the owner and the attacker by more than 40% across various app-contexts.

To measure the performance of the developed model using specific app interactions, such as in the case of Chrome, allows us to analyze the distribution of inferences for both the legitimate owner and potential attackers over a session lasting for two minutes, as illustrated in Fig. 6b. The choice of Chrome app, a widely used browser, enables us to observe user behavior in a web-browsing context, which is representative of real-world smartphone usage. Fig. 6b demonstrates that the model's performance was robust within this context. The inference accuracy for the legitimate owner consistently reached 98%, with no inference falling below 82%. In contrast, the attacker's inference varied between 6% and 13%. This targeted exploration of this widely used app not only offers insights into typical user behavior patterns but also show the model's ability to distinguish between the owner and potential attackers within a specific app environment. This focused analysis on the Chrome app aligns with our goal of understanding behavioral biometrics in real-world, commonly used applications, providing valuable insights into the practical effectiveness of our proposed framework.

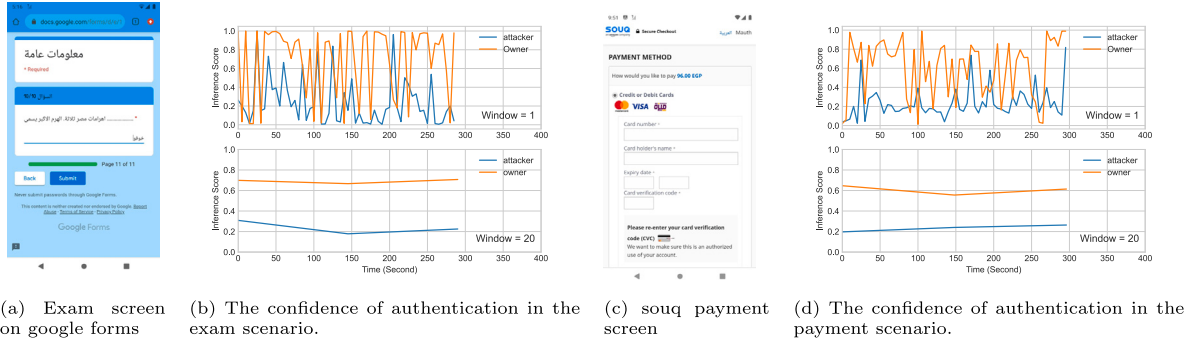


Fig. 7. Context analysis over two high risk apps.

6.8. High risk study

To validate the concept of behavioral authentication in a real world scenario, we asked a participant to use two different apps classified as high risk ones while the trained authentication model works in the background [43]. In the first scenario, as shown in Fig. 7(a), the participant solved an exam consisting of ten questions, with 58 touch actions recorded during the exam, of which 46 were taps due to the app's context requiring more taps than slides. Another participant played the role of an attacker and attempted to solve the same exam, resulting in 53 touch actions, 50 of which were taps. The authentication model's performance, as depicted in Fig. 7(b), showed a significantly higher confidence level for the owner compared to the attacker, demonstrating the model's efficiency in distinguishing between legitimate users and imposters.

Similarly, we did the same scenario for a shopping app (i.e., Souq) with a payment screen as shown in Fig. 7(c). The owner performed 59 touches to finish the payment process, while the attacker performed 78 touches. Fig. 7(d) shows a higher level of confidence in authenticating the owner compared to the attacker. Remarkably, the implicit authentication model that was trained by the owner data with only 577 touch actions during approximately 27 min, maintained an inference accuracy of over 60% for the owner, while the imposter's inference accuracy remained below 30% during the period of payment.

This real-world validation demonstrates the effectiveness of the trained implicit authentication model in differentiating between legitimate users and attackers, showcasing its potential for secure and seamless behavioral authentication in real-world usage scenarios.

7. Discussion

The presented behavioral biometric authentication framework, B2auth, has shown promising results in differentiating between the owner and imposter, providing a usable and secure user authentication experience. Through the aforementioned comprehensive evaluation, we have gained valuable insights into the B2auth performance, usability, and potential implications for real-world deployment. The validation of B2auth was conducted using real-world scenarios. The results revealed the model's ability to distinguish between the legitimate user and attacker. For instance, during the exam and payment scenario apps, the model shows a significant higher confidence levels for the owner compared to the attacker, indicating accurate and reliable authentication.

The usability evaluation indicated positive user feedback, showing a favorable user experience with B2auth framework. The authentication procedure was easy and comfortable for participants, showing its seamless integration into normal smartphone usage. The system's real-time operation on the device contributed to quick and efficient verification, eliminating the need for internet connectivity and reducing latency. Users appreciated the minimal touch actions required, contributing to an overall positive user experience. These findings highlight the model's potential to provide a non-intrusive and frictionless authentication experience for users.

B2auth demonstrated promising security characteristics, effectively identifying legitimate users while minimizing false acceptance rates. The framework's performance was resilient against potential spoofing attacks, as attackers failed to replicate genuine users' behavioral patterns accurately as we did in the risk study in Section 6.8. This resilience against attacks fosters confidence in the system's ability to protect sensitive data and secure access to high-risk applications.

The cloud-based model generation and on-device inference allowed B2auth to perform offline predictions, conserving device resources and ensuring continuous operation even in resource-constrained environments. The implementation of the B2auth framework demonstrated resource efficiency, with minimal computational overhead on the mobile device.

7.1. Comparative analysis

In this section we conducted a comparative analysis between the model's performance during the training phase and its performance during deployment. Our goal is to have a model that not only performs well during training but also generalizes

Table 8
Comparison Summary of AUC and EER Values during training and deployment.

Metric	Training		Deployment	
	Tap	Slide	Tap	Slide
AUC	0.78	0.87	0.79	0.79
EER	0.29	0.23	0.29	0.30

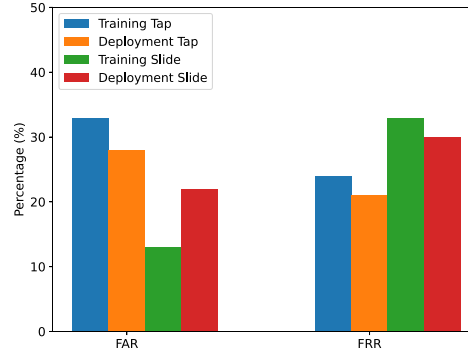


Fig. 8. False accept rate and false reject rate during training and deployment.

Table 9
Performance comparison of the trained version of B2auth with benchmark models.

Model	Method	AUC (Slide)	AUC (Tap)	EER (Slide)	EER (Tap)
BehavePassDB [44]	LSTM RNN	0.732	0.645	–	–
MMauth [28]	DeSVDD	–	–	0.149	0.149
IncreAuth [29]	GBDTNN	–	–	0.087	0.087
B2auth (Proposed)		0.869	0.776	0.238	0.294

effectively to real-world scenarios. Table 8 presents a comparison between the performance of Tap and Slide models during the training and deployment. The tap model accuracy remained consistent for training and deployment where the AUC was 0.78 and 0.79 respectively with an EER of 0.29 for both phases. For the slide model there was a variation between the performance of the models, where the performance of the slide model decreased during the deployment scenario by almost 8%. This reflects the variations in user behavior and device usage, influenced the slide model's performance more than the tap model. This aligns with expectations, given the nature of slide interactions.

In regards to the error rate analysis, Fig. 8 shows the FAR and FRR for both tap and slide models across training and deployment phases. The tap model maintains its stability in FAR and FRR, highlighting its reliability in distinguishing between owner and illegitimate users. In the slide model, FAR has increased by 9%, but FRR improved during deployment. Despite the minor variations in slide model, both models maintain practical applicability for diverse real-world authentication scenarios.

7.2. Comparison with the previous work

Most previous work models were evaluated using offline data collected during studies, while B2auth were evaluated in a deployment operational scenario adding a significant effort in the behavioral biometric authentication field. This real-world deployment provides insights into B2auth's practical effectiveness in a live operational context.

In this section we present several distinct features of B2auth in compare with other benchmark models. Due to the lack of operational evaluation scenario in the field of behavioral biometric authentication, we compare the training evaluation results of the B2auth with benchmark models shown in Table 9. Also, B2auth focus on using touchscreen data only while the majority of other models leverage different sources of data such as sensors.

Table 9 shows the comparison with the touch results only using the AUC and EER for both slide and tap authentication. B2auth model achieves an AUC performance of 0.869 for slides and 0.776 for taps, surpassing the performance of BehavePassDB's which is 0.732 for slides and 0.645 for taps. MMauth's and IncreAuth's used EER as an evaluation measure to their developed models. Although these models present lower EER with 0.149 for slides and taps, and 0.087 for slides and taps, respectively. B2auth manage to achieve an EER of 0.238 for slides and 0.294 for taps with a comparatively small number of touches (i.e., approximately 362 taps and 215 slides on average). Also, B2auth is designed for efficiency and real-world deployment, contributing to its practicality and usability in operational scenarios.

Also these models contains a larger and more complex neural networks architecture, while B2auth designed using a simple MLP neural network that comprises three hidden layers prioritizes efficiency without compromising performance. The decision to keep the MLP network intentionally simple was motivated by the need to minimize on-device inference costs, especially when running concurrently with other applications.

8. Conclusion and future work

In this paper we have presented B2auth a behavioral biometric authentication framework. Its architecture contains two main components, enrolment phase and verification phase. The enrolment phase contained, dataset construction and labeling, training, validation and testing, model generation, and model deployment. In the verification phase, the generated model was used to authenticate the user. Then we conducted a set of experiments to validate the applicability of using implicit authentication in a real world scenario. The best results was achieved suing Tap+Slide touches during a frequency window of 20 samples with AUC of 80% and EER of 29%. The real-world validation showcases the model effectiveness in differentiating between legitimate users and attackers. As behavioral biometrics continue to evolve, B2auth serves as a promising step towards achieving seamless and reliable authentication in modern mobile computing environments.

While B2auth has exhibited promising performance, there are avenues for further improvement and exploration. Future research could conducted to enhance feature extraction methods to capturing additional behavioral traits for multi-modal authentication. Optimization of the model training process and exploration more sophisticated deep learning architectures could lead to a higher accuracy and robustness. Additionally, extending the framework to support other biometric traits, such as gait-based or voice-based authentication, could present an opportunity for more comprehensive and secure user verification. We are also planning to extend our study to include different devices and deployed it on users' own phones for a longer period of time to measure the aging of training data.

CRedit authorship contribution statement

Ahmed Mahfouz: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Ahmed Hamdy:** Funding acquisition, Investigation, Project administration, Resources, Supervision, Visualization, Writing – review & editing. **Mohamed Alaa Eldin:** Data curation, Formal analysis, Methodology, Resources, Software, Validation, Visualization, Writing – original draft. **Tarek M. Mahmoud:** Data curation, Formal analysis, Methodology, Resources, Software, Validation, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Study procedure

The study procedure will be as follows [45,46]:

1. Preparing the smartphones instrumented with the B2auth framework.
2. Make an advertisement campaign for participant recruitment.
3. Make a registration form to collect demographics.
4. Invite the selected participants for an interview.
5. Explain the study in details to each subject.
6. Assign a unique anonymous ID for each invited participant.
 - (a) Give the smartphone to the participant and ask him to do a set of tasks based on the developed scenarios for training purpose.
 - (b) Upload the collected features to GCP for training and generating the AI authentication model.
 - (c) Download the trained model and change the phone setting to be in the authentication mode.
 - (d) Give the smartphone again to the participant and ask him to do a set of tasks based on the developed scenarios for authentication purpose.

Appendix B. Enrolment scenario

A set of common daily activities that participants performed in the field study:

1. Enter either pattern or PIN for 5 times depending on the credential that has been used in her/his phone.
2. Enter the Facebook Login credentials and make a post in the wall page, browse the page and interact with likes and comments then logout.
3. Open Contacts and add a new one then make a phone call using Dailer app.
4. Take a picture “nature” using the Camera app and share it on social media.
5. Send a Message using Whatsapp or Messenger to his phone number.
6. Change the phone Settings, network, notifications, brightness level, wallpaper.
7. Open the play store and install a new app.
8. Search for the price of note 10 lite device.
9. Open google map and look for location and direction for the train station.
10. Play one game of Fruit Ninja and candy crush.
11. Open eMail and check inbox, read the new emails, and reply to the email sent by the study.

Appendix C. Selected features

C.1. Slide features

- **X, Y Co-ordinates:** downX, downY, upX, upY
- **Touch Pressure:** downTouchPressure, upTouchPressure, avgTouchPressure, minTouchPressure, maxTouchPressure
- **Touch Area:** downTouchArea, upTouchArea, avgTouchArea, minTouchArea, maxTouchArea
- **Velocity:** avgVelocity, maxVelocity
- **Length, Displacement, Duration, Curvature, Acceleration**

C.2. Tap features

- **X, Y Co-ordinates:** downX, downY, upX, upY
- **Touch Pressure:** downTouchPressure, upTouchPressure, avgTouchPressure, minTouchPressure, maxTouchPressure
- **Duration**

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